

Class 10: Structural Biochemistry

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Table of contents

First look at the data	1
Analysing with Molstar	4
Reading pdb data into R	9
Pdb visualization in R	10
Predicting functional motions of a single structure	10
PCA Analysis	16

First look at the data

A key point is that we have a very, very small coverage of known proteins

```
pdb <- read.csv("Data Export Summary.csv", row.names=1)
convert <- function(y){
  return (as.numeric(sub(",", " ", x=pdb[[y]]))) #as.numeric() converts str to int
} # double bracket [[]] is used to handle for columns given that it accepts variables
for (i in colnames(pdb)){
  pdb[[i]] = convert(i) #I made the convert function
}
#sum(pdb$X.ray)
pdb
```

	X.ray	EM	NMR	Integrative	Multiple.methods	Neutron
Protein (only)	180758	23111	12813	348	229	84
Protein/Oligosaccharide	10488	3741	34	8	11	1
Protein/NA	9205	6751	287	26	8	0
Nucleic acid (only)	3154	250	1578	3	15	3
Other	178	27	35	4	0	0

Oligosaccharide (only)	11	0	6	0	1	0
	Other	Total				
Protein (only)	32	217375				
Protein/Oligosaccharide	0	14283				
Protein/NA	0	16277				
Nucleic acid (only)	1	5004				
Other	0	244				
Oligosaccharide (only)	4	22				

Q1. 93.9% of all structures in the pdb dataset were solved by X ray or EM

```
(sum(pdb$X.ray) + sum(pdb$EM))/sum(pdb$Total) # sum gives a vec total: arithmetic
```

```
[1] 0.9386623
```

Q2. 98.0% of structures in the PDB are proteins

```
(pdb["Protein (only)","Total"]+pdb["Protein/Oligosaccharide","Total"]+pdb["Protein/NA",
```

```
[1] 0.9791868
```

Q3. There are 1227 HIV-1 protease structures in the PDB database

Analysing with Molstar

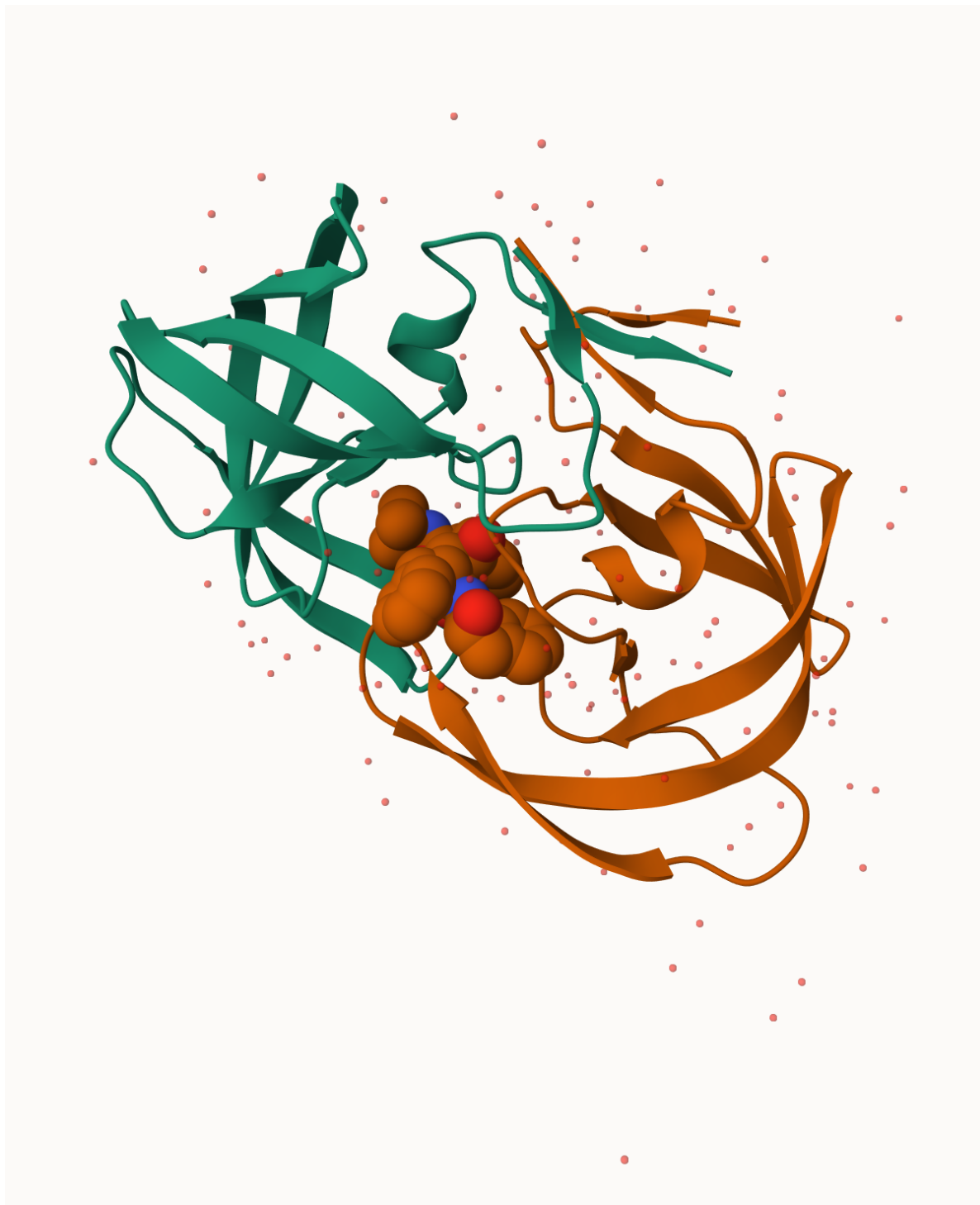


Figure 1: The HIV-protease enzyme is a homodimer with two chains

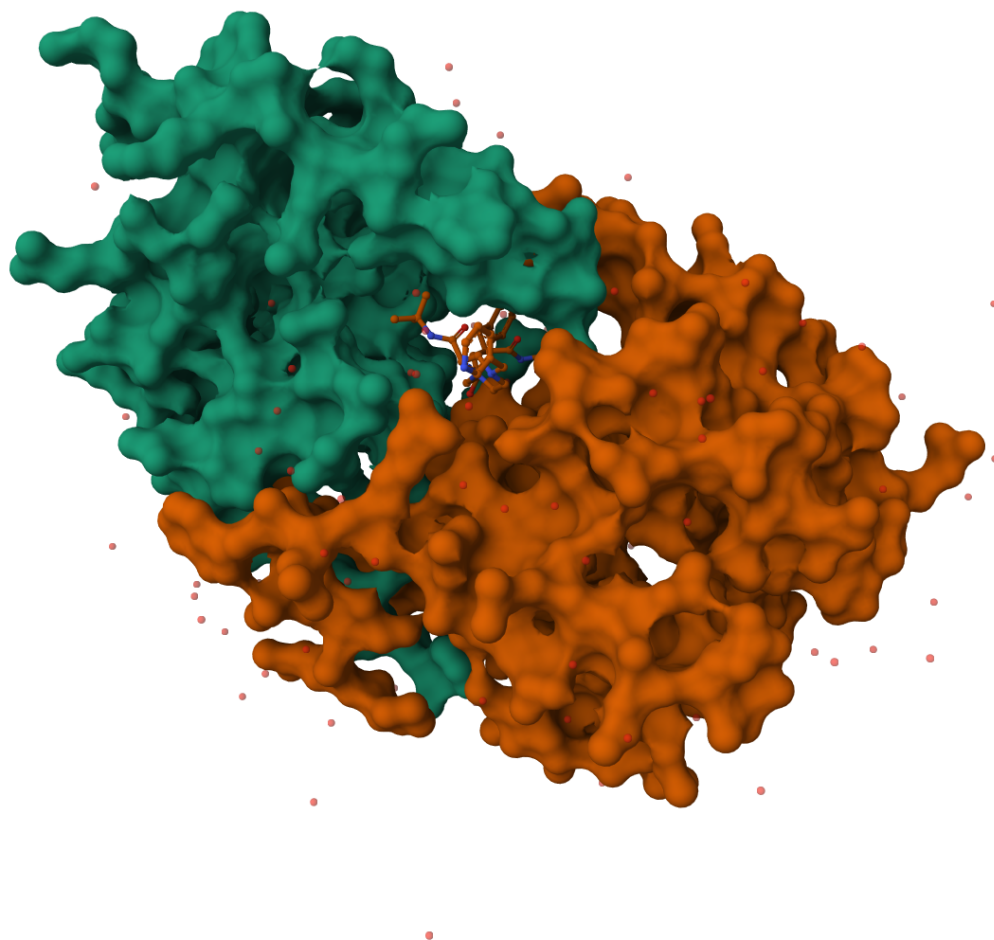


Figure 2: This is the HIV-protease enzyme with polymers in the molecular surface representation form

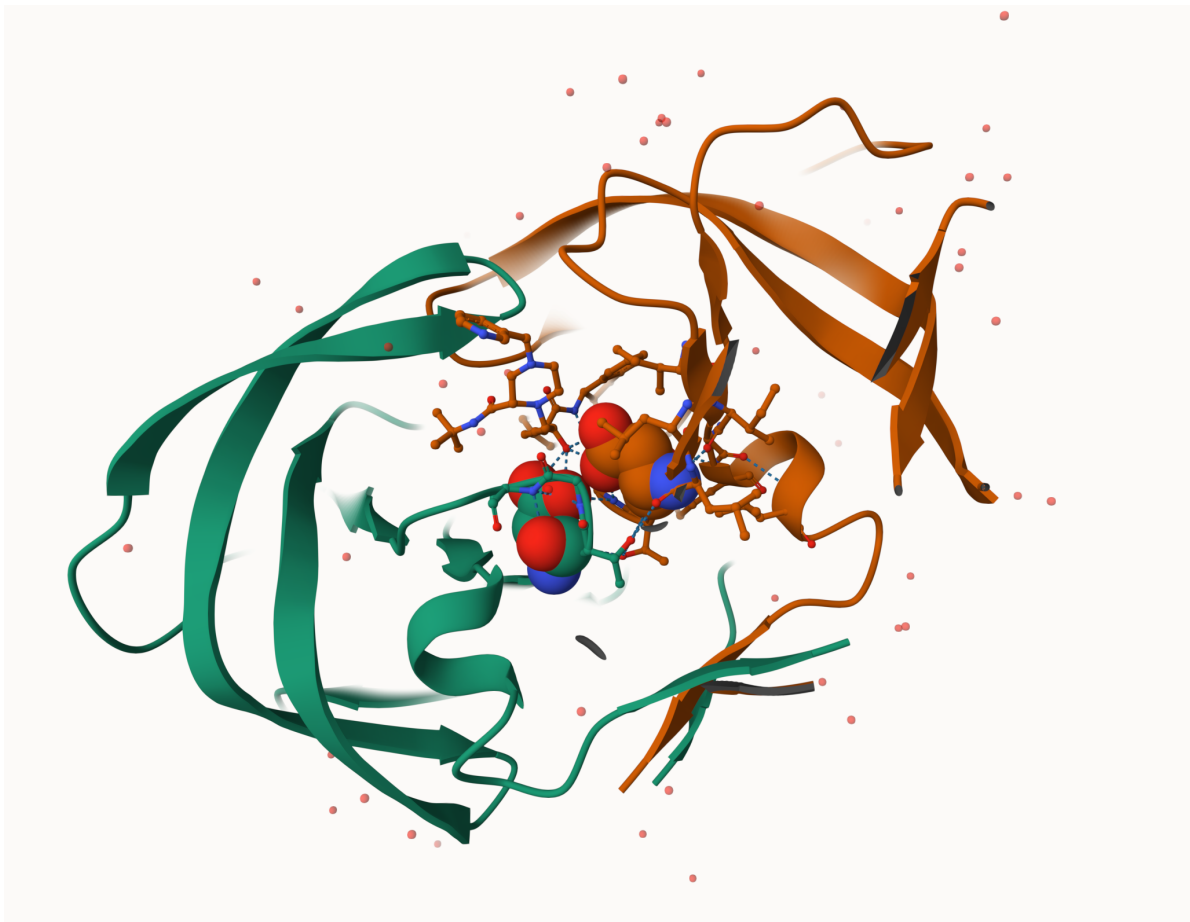


Figure 3: This is the HIV-protease enzyme highlighting the two key aspartate residues ASP-25 on both chains

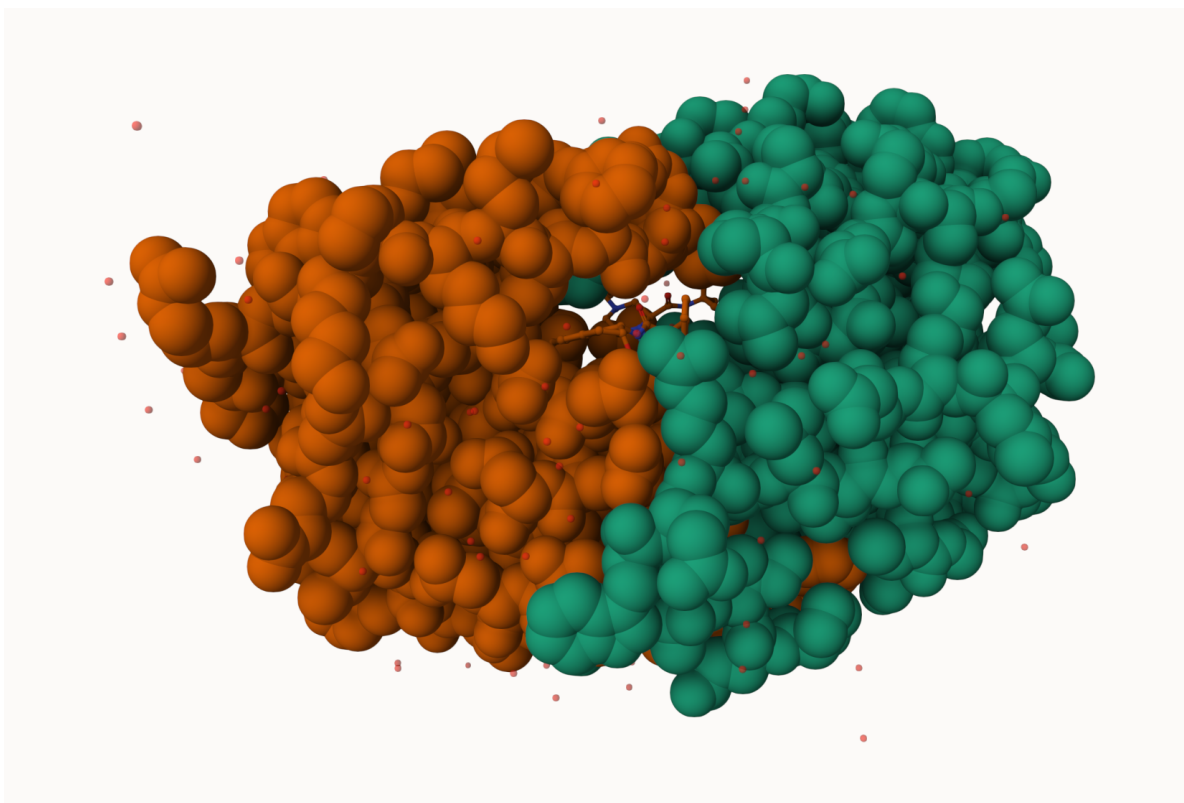


Figure 4: This is the HIV-protease enzyme with space-fill representation of protein polymers, highlighting ligand position in the protein.



Figure 5: The water molecule in question is placed into space-fill view and highlighted along with the two ASP-25 residues

Q4. We only see one atom per water molecule because at this scale of view in terms of Angstrom, our resolution is too large to visualize the hydrogen atoms, which are extremely small

Q5. This water molecule is number 308, and is visualised in the figure above

Q6. This figure is made above

Reading pdb data into R

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
Q7. 198 amino acid residues
```

```
Q8. HOH
```

```
Q9. 2 Chains: A and B
```

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

Pdb visualization in R

```
library(bio3dview)
library(NGLViewerR)

#view.pdb(pdb) |> # allows viewing of a pdb file
#setSpin()
```

```
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
#view.pdb(pdb, cols=c("navy","teal"),
  # highlight = sele,
  #highlight.style = "spacefill") |>
#setRock()
```

Predicting functional motions of a single structure

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

Call: read.pdb(file = "6s36")

Total Models#: 1

Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)

Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:

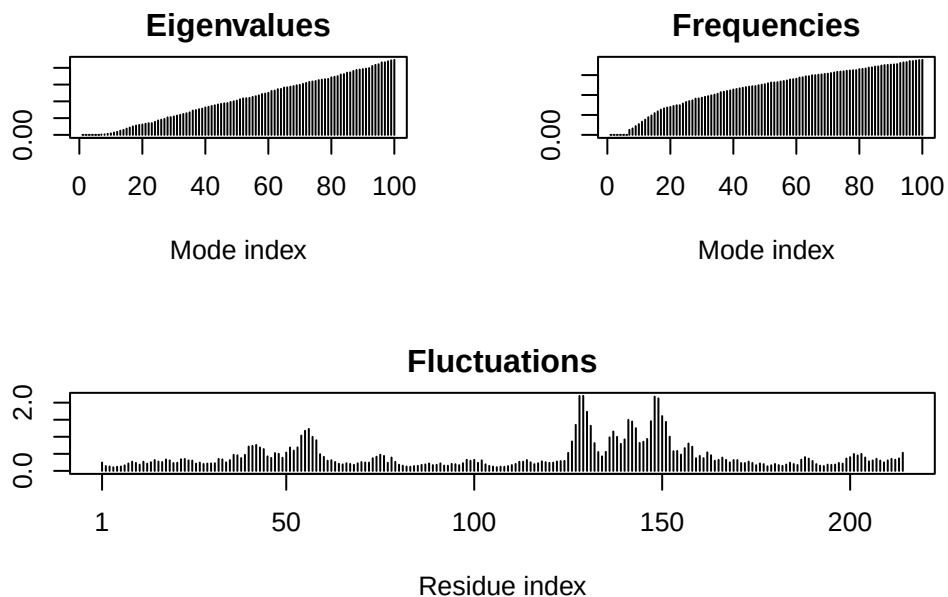
```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLDGFPRITPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
# Perform flexibility prediction  
m <- nma(adk)
```

```
Building Hessian...      Done in 0.013 seconds.  
Diagonalizing Hessian... Done in 0.295 seconds.
```

```
plot(m)
```



```
mktrj(m, file="adk_m7.pdb")  
#view.nma(m, pdb=adk)
```

Q10. msa is the only package above found only on bioconductor and not CRAN

Q11. bio3dview is not found on either bioconductor or CRAN

Q12. This is true

```
library(bio3d)
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
#blast <- blast.pdb(aa)
aa
```

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTPALIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

+ attr: id, ali, call

Q13. This sequence is 214 amino acids long

```
hits <- NULL
hits$pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A', '6HAM_A')
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download
```

```
|
|
| 0%
|====
| 8%
|=====
| 15%
|=====
| 23%
|=====
| 31%
|=====
| 38%
|=====
| 46%
|=====
| 54%
|=====
| 62%
|=====
| 69%
|=====
| 77%
|=====
| 85%
|=====
| 92%
|=====
| 100%
```

```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
```

```

pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...

```

Extracting sequences

```

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10  name: pdbs/split_chain/6HAM_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdbs/split_chain/4K46_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13  name: pdbs/split_chain/4PZL_A.pdb

```

```
library(bio3dview)
```

```
#view.pdbs(pdbs)
```

```
ids <- basename.pdb(pdbs$id)
```

```
anno <- pdb.annotate(ids)
```

```
unique(anno$source)
```

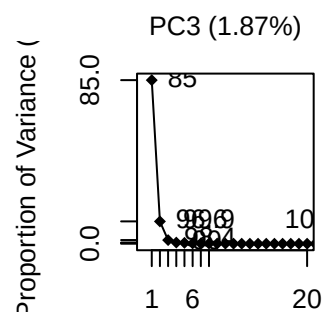
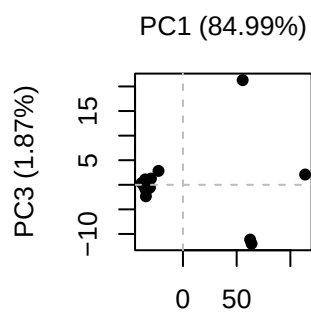
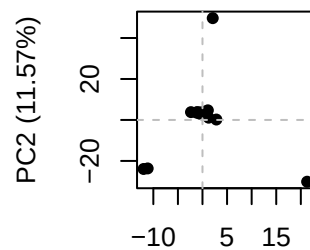
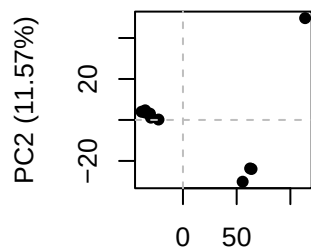
```
[1] "Escherichia coli"  
[2] "Escherichia coli K-12"  
[3] "Escherichia coli O139:H28 str. E24377A"  
[4] "Escherichia coli str. K-12 substr. MDS42"  
[5] "Photobacterium profundum"  
[6] "Burkholderia pseudomallei 1710b"  
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

PCA Analysis

```
# Perform PCA
```

```
pc.xray <- pca(pdbs)
```

```
plot(pc.xray)
```

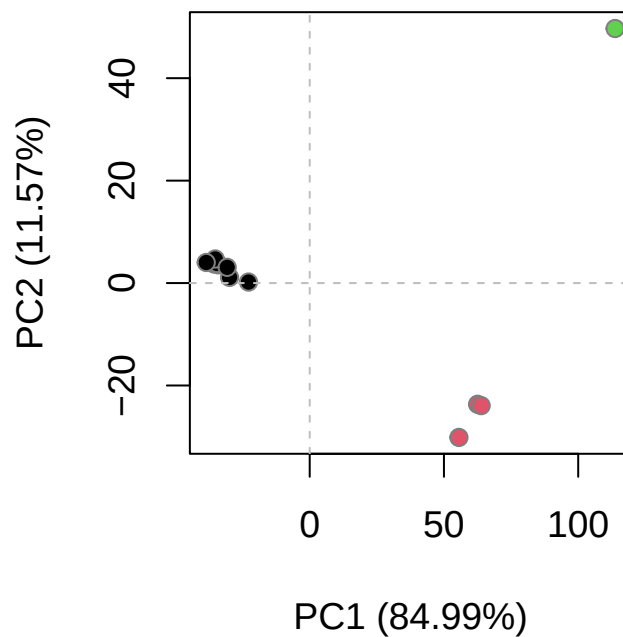



```
# Visualize first principal component
# Calculate RMSD
rd <- rmsd(pdbbs)
```

Warning in rmsd(pdbbs): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                  PC2=pc.xray$z[,2],
                  col=as.factor(grps.rd),
```

```

ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p

```

